

Optics Letters

Fiber-optic joint time and frequency transfer with the same wavelength

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Received 2 October 2019; revised 27 November 2019; accepted 27 November 2019; posted 27 November 2019 (Doc. ID 379338); published 24 December 2019

Optical fiber links have proven their powerful ability to transfer the ultrastable clock signals over a national and continental metrological optical network. In this Letter, we propose and demonstrate a new scheme that time and frequency signals are modulated on the same laser wavelength with phase and intensity modulation by an independent electro-optic modulator, respectively. Different wavelengths are used in two opposite directions to prevent Rayleigh scattering. The noise changes caused by the fluctuations of the transfer links are actively canceled at the local site by optical delay lines. Radio frequency with 1 GHz and time signal with one pulse per second (1 PPS) are transmitted over 110 km fiber spools. The experimental results demonstrate that frequency instabilities of $1.7\text{E} - 14$ at 1 s and $5.9\text{E} - 17$ at 10^4 s and time interval transfer of 1 PPS signal with sub-ps stability after 1000 s are obtained. This novel scheme offers advantages with respect to reduce the channel in the fiber network, and can keep the time and frequency signal free from interference with each other. © 2019 Optical Society of America

<https://doi.org/10.1364/OL.45.000208>

The synchronization of time and frequency between remote locations is very important in many areas of human activity involving Global Navigation Satellite System (GNSS) [1] as well as very long baseline interferometry (VLBI) [2] and some commercial applications (e.g., metrology and timescale development). With the advances in modern atom clocks, such as cesium fountains, hydrogen masers, and optical clocks, the frequency instability is better than $1\text{E} - 13$ at 1 s for microwave signal [3,4]. However, the comparison of the radio frequency (RF) signal using satellite-based techniques is limited at a level around $1.0\text{E} - 15$ over one day. Optical fiber has provided a rapidly emerging alternative for long-distance time and frequency dissemination, offering much better accuracy.

Joint time and frequency transfer in optical fiber has been widely discussed, and there are several kinds of schemes. One classical method for the purpose is transferring time and frequency signals in two adjacent wavelengths [5]. More specifically, time and frequency signals are modulated on the optical carriers from two lasers that have different wavelengths of about 0.8 nm by intensity modulation (IM), and two dense wavelength division multiplexers (DWDM) are used to combine and separate the modulated laser at the local and remote sites (RSs). The dissemination stability is improved to $7.0\text{E} - 15/1$ s and $4.5\text{E} - 19/1$ day at the distance of 80 km, and time synchronization at the 50 ps precision level was also demonstrated [6]. However, since the communication channels in metropolitan network are scarce resources, this method may occupy an additional communication channel. To get around this issue, scientists have proposed another scheme, in which the sinusoidal frequency is converted into a square signal, and the time signal [1 pulse per second (PPS)] is incorporated by triggering a violation made on the square wave by an embedder [7]. This scheme can solve the above problem, but the converting process of the sinusoidal frequency signal may introduce electronics noise from digital circuits. Those noises will cause a deterioration of frequency stability and make the short-term stability worse than $2.0\text{E} - 14/10$ s [8]. In the third scenario, a mode-locked laser is employed to transfer the time signal with IM, where time deviation 300 fs at 5 s is obtained and optical and microwave frequency at a level better than $1.0\text{E} - 17$ are promising [9]. However, the technique is complicated because optical frequency combs are needed. In this Letter, we propose a simple scheme to jointly transfer the time and microwave frequency signal to the RS with the same wavelength, whose major superiority is reducing the channel number in the fiber network. At the local site (LS), the time signal is phase modulated, and the frequency signal is intensity modulated on the same carrier. At the RS, the transferred optical carrier is split into two beams, and time and frequency signals are recovered by coherent and incoherent demodulation, respectively. The noise caused by

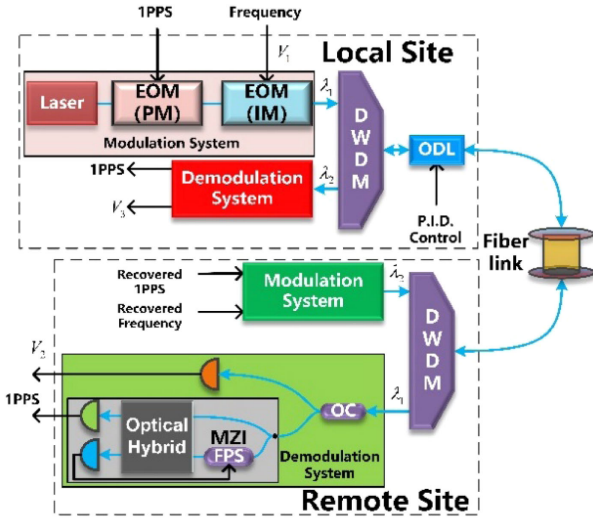


Fig. 1. Schematic diagram of the proposed time and frequency transfer system. EOM, electro-optical modulator; DWDM, dense wavelength division multiplexer; OC, optical coupler; ODL, optical delay line; FPS, fiber phase shifter.

the fiber link is compensated by the optical delay line (ODL). Time and frequency transfer over 110 km fiber spools based on the proposed scheme is demonstrated. Relative stabilities of $1.7\text{E} - 14/1\text{ s}$ and $5.9\text{E} - 17/10^4\text{ s}$ for the frequency and time interval for 1 PPS with sub-ps stability after 1000s are obtained. Furthermore, time signals in two locations can be synchronized with the method described in [7], and they can also be calibrated with Universal Time Coordinated (UTC).

Figure 1 illustrates the diagram of the proposed scheme. At the LS, the reference time and frequency are modulated by the modulation system, which contains the laser source and two electro-optical modulators (EOM). The modulation modes of two EOMs are different, one of which is based on IM and the other one is based on phase modulation (PM). The photoelectric field of the optical carrier from the laser source can be expressed as a simple E function,

$$E_1 \propto \exp i (\omega_c t + \varphi_0), \quad (1)$$

where ω_c is the optical frequency of the laser source and φ_0 is the initial phase.

The pulse amplitude is chosen in such a way that the π -phase step is imprinted at the first EOM output. After that, we obtain the photoelectric field

$$E_2 \propto \exp i [\omega_c t + \varphi_0 + \varphi(t)], \quad (2)$$

where $\varphi(t)$ is the encoded single pulse.

The sine frequency signal can be expressed as

$$V_1 \propto \sin (\omega_0 t + \varphi'_0), \quad (3)$$

where ω_0 is the frequency of sine signal and φ'_0 is the initial phase. The frequency signal is intensity modulated on the optical carrier by the second EOM. The light intensity after the second EOM is proportional to

$$I_1 \propto [1 + \alpha \sin (\omega_0 t + \varphi'_0)] |\exp i [\omega_c t + \varphi_0 + \varphi(t)]|^2, \quad (4)$$

where α represents modulation depth.

The optical carrier after the modulation system in the LS passes through the DWDM and injects into the ODL. The ODL contains a 10 km temperature-controlled fiber-optic spool (bandwidth of several Hertz) and a short fiber wrapped around piezoelectrical transducer (PZT, bandwidth 2kHz) [10] in order to cancel out the slow and fast propagation delay caused by the fiber link. After the ODL, the forward signal passes the fiber link and arrives the RS, and its light intensity can be expressed as

$$I_2 \propto [1 + \alpha \sin (\omega_0 t + \varphi'_0 + \varphi'_{\text{ODL}} + \varphi'_1)] \times |\exp i [\omega_c t + \varphi_0 + \varphi_{\text{ODL}} + \varphi_1 + \varphi(t)]|^2, \quad (5)$$

where φ'_1 is the fiber link propagation delay from LS to RS for the RF signal and φ'_{ODL} is the phase shift caused by ODL. φ_1 and φ_{ODL} are the phase shifts caused by fiber link and ODL for optical frequency, respectively.

The optical carrier arriving at the RS can be recovered by the demodulation system, which is shown in Fig. 1 with a green block diagram. The light signal at the end of the fiber link is divided into two parts by a 50/50 optical coupler (OC), One part of which is inputted into a high-bandwidth photodetector (orange, Newport1611, bandwidth 30 kHz–1 GHz) and converted back to the electrical domain by direct detection (DD) of the intensity of received light. Ideally, pure PM will not affect the intensity of the light signal [11]. After the detector, we obtain the frequency signal,

$$V_2 \propto \sin (\omega_0 t + \varphi'_0 + \varphi'_{\text{ODL}} + \varphi'_1). \quad (6)$$

Then, the frequency signal is divided into two parts by a power splitter (shown in Fig. 2). One is remodulated back to the LS, the other is for the RS users.

The other part of the light signal from the OC is demodulated by a Mach-Zehnder interferometer (MZI), which is shown in gray block diagram, and the 1 PPS pulse will produce by contrast two constructive interferences after recombination by the MZI [12–14]. The MZI contains an optical hybrid, two photodetectors, and one fiber phase shifter (FPS, General Photonics, FPS-003-35-SS). This MZI has unbalanced paths and acts as delay line interferometer. The path difference of two arms of the MZI equals as τ (τ is the width of the 1 PPS signal).

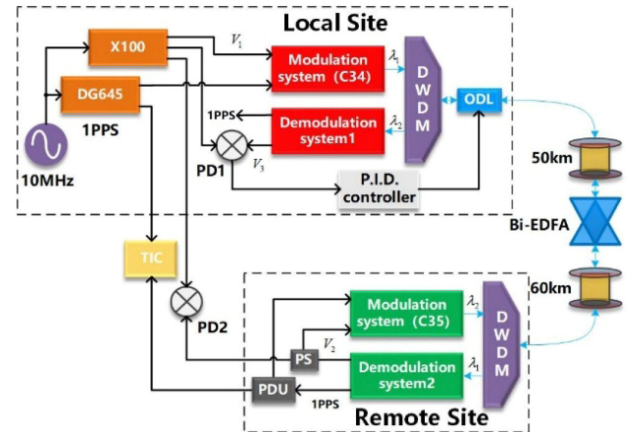


Fig. 2. Experimental setup of the time and frequency transfer. PD, phase discriminator; PS, power splitter; PDU, pulse distribution unit.

The photoelectric fields of two beams inside the MZI can be expressed as

$$E_{\text{MZI},1} \propto \sqrt{1 + \alpha \cos(\omega_0 t + \xi)} \exp i [\omega_c t + \varphi(t)], \quad (7)$$

and

$$E_{\text{MZI},2} \propto \sqrt{1 + \alpha \cos[\omega_0(t + \tau) + \xi]} \times \exp i [\omega_c(t + \tau) + \varphi(t + \tau) + \Delta\theta], \quad (8)$$

where

$$\xi = \varphi'_0 + \varphi'_{\text{ODL}} + \varphi'_1, \quad (9)$$

and $\Delta\theta$ is the additional phase difference caused by thermal and vibration.

After the optical hybrid, light intensity of two outputs can be described as

$$I_1 \propto 2 + \alpha \cos(\omega_0 t + \xi) + \alpha \cos[\omega_0(t + \tau) + \xi] + 2\sqrt{[1 + \alpha \cos[\omega_0(t + \tau) + \xi]] \cdot [1 + \alpha \cos(\omega_0 t + \xi)]} \times \cos[\omega_c \tau + \Delta\theta + \varphi(t + \tau) - \varphi(t)], \quad (10)$$

$$I_Q \propto 2 + \alpha \cos(\omega_0 t) + \alpha \cos[\omega_0(t + \tau)] + 2\sqrt{[1 + \alpha \cos[\omega_0(t + \tau) + \xi]] \cdot [1 + \alpha \cos(\omega_0 t + \xi)]} \times \sin[\omega_c \tau + \Delta\theta + \varphi(t + \tau) - \varphi(t)]. \quad (11)$$

I_Q can be detected by a photodetector (blue, Thorlabs, PDB140C) and passes a low-pass filter (bandwidth DC-1.9 MHz), from which we just get the error signal. The converted electrical signal can be described as

$$V_Q \propto A + \sin(\omega_c \tau + \Delta\theta), \quad (12)$$

where A is a constant.

One circuit controls the FPS and makes $\sin(\omega_c \tau + \Delta\theta)$ equal to zero. Thus, the optical phase difference can be stabilized, and I_1 can reach maximum pulse amplitude.

I_1 can be converted to the electrical domain by DD, using a low-bandwidth detector (green, Newport 1811, bandwidth DC-125 MHz). This detector can act as a low-pass filter, and it can filtrate the frequency signal. The time signal is recovered only after detector 1811 and input into a pulse distribution unit (PDU, Timetech No10188), whose outputs include the return 1 PPS and signal for users in the RS.

The experimental setup is shown in Fig. 2. A rubidium atomic clock generates a 10 MHz microwave signal and is split into two branches by a power splitter. One branch passes through a phase-lock frequency multiplier to obtain a 1 GHz as the standard RF signal V_1 . Meanwhile, the 1 PPS time signal is produced by a time generator DG645, which is locked by the other frequency branch. Time and frequency signals are modulated on the laser in the modulation system shown in Fig. 1 as the red block diagram. The optical carrier in LS has the wavelength $\lambda_1 = 1550.1$ nm (C34) and in RS has the wavelength $\lambda_2 = 1549.3$ nm (C35). Different wavelengths are used to prevent Rayleigh scattering in the fiber spools and bidirectional erbium-doped fiber amplifier module (Bi-EDFA), and

the influence of bidirectional asymmetry can be neglected when the wavelength gap is small enough [15]. Two DWDMs are applied to separate the forward and backward optical carriers at the LS and the RS. The fiber link contains a 50 km and a 60 km fiber spool, and between the two spools is a Bi-EDFA to amplify the amplitude of optical signals in both directions. At the RS, the signal carried on λ_1 is demodulated as shown in Fig. 1 and transmitted back along the same fiber link with the wavelength λ_2 . The round trip signal from the RS at the LS is demodulated by the same structure. The recovered frequency signal in LS and standard RF signal are input to a phase discriminator (PD1, HMC 439) to get the error signal [16].

After the process of proportional-integral-derivative (PID), we apply this error signal to drive the ODL, making it actuate a phase correction of $\varphi'_{\text{ODL}} = -\varphi'_1$. That means the whole fiber link is stabilized. That is to say, we can obtain stabilized time and frequency signals in the RS [17]. The standard RF in the LS and the converted signal in the RS are input to the second phase discriminator (PD2) for the performance evaluation of frequency transfer. A time interval counter (TIC, GT668) is used to continuously measure the time delay between the converted 1 PPS in the RS and that from the output of DG645.

Figure 3 shows the time delay drift and the frequency phase drift in the time domain over 110 km fiber link. The phase drift in open loop is mainly due to the temperature change, and the range of delay variation from fiber link is nearly 1800 ps. It can be clearly observed that the time delay and frequency phase drift can fit well except that there is a tiny difference from 35,000 s to 52,000 s in open loop. We estimate this difference is due to the dead zone of the phase discriminator because of drastic phase drift in the open loop. For the case of the closed loop, the fiber link is stabilized, and phase drift is limited in a small range; a dead zone does not exist. The blue and magenta lines show the time and phase drift for the case of closed loop. The peak-to-peak value of the time delay drift is reduced to less than 120 ps, and the phase drift is reduced to less than 0.2 ps.

Figure 4(a) shows the measured time deviation (TDEV) of the proposed scheme over 110 km fiber link. The red line is the result when the fiber link is running freely. It has short-term stability of $2.7\text{E} - 11/1$ s, and long-term stability worsens seriously because of temperature fluctuation. The dark line is the result when the phase noise is compensated, the users in RS can get a time signal with the relative stability of $1.6\text{E} - 11/1$ s and $9.1\text{E} - 13/10^4$ s. There is a great improvement at averaging

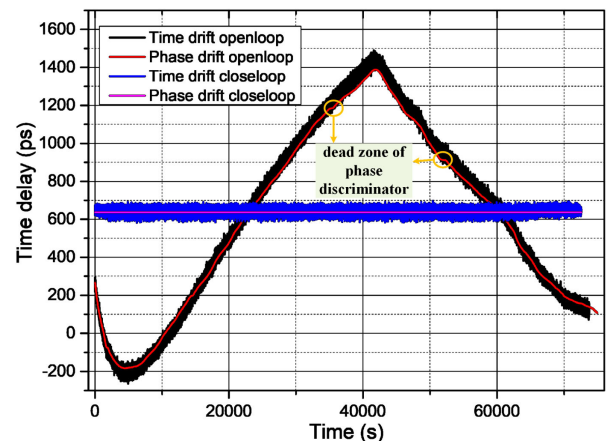


Fig. 3. Time and phase drift between the RS and the LS.

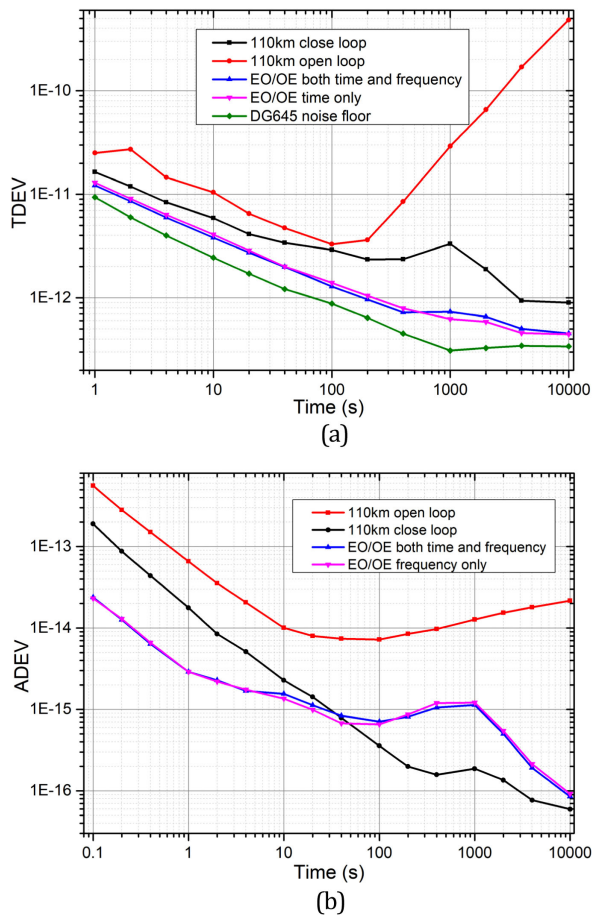


Fig. 4. (a) Measured time deviation (TDEV) and (b) Allan deviation (ADEV) of the experiment.

time of 100–10,000 s when comparing the performance of the closed loop with that of the open loop. Blue and magenta lines represent the TDEV noise from the electric–optic–electric (EO/OE) process when both time and frequency are modulated and only time is modulated, respectively. This EO/OE noise refers to free-running single trip and is obtained by connecting the LS and RS with a short fiber. The results from blue and magenta lines show that 1 GHz frequency signal with IM has little impact on single pulse after the low-pass detector at the level of 10 ps. At averaging time of 1000 s, a periodic oscillation is obvious. The period of this oscillation coincides with that of our air conditioner, which can also be observed in frequency stability. The other limit of TDEV comes from the noise floor of DG645 (green line, by measuring the time jitter between two outputs of DG645), which is about $9.4\text{E} - 12/1\text{ s}$ and $3.2\text{E} - 13/10,000\text{ s}$. The relative stability (Allan deviation, ADEV) of frequency transfer with a 5 Hz measurement bandwidth is shown in Fig. 4(b). For the case of the closed loop over 110 km fiber link, relative stability is $1.7\text{E} - 14/1\text{ s}$ and $5.9\text{E} - 17/10^4\text{ s}$. Similarly, the EO/OE noise in two cases shown in Fig. 4(b) indicates that the noise induced by residual

amplitude modulation (RAM) [11] has a negligible impact on frequency stability at the level of femtoseconds.

In summary, we have proposed and demonstrated a fiber-based scheme that can transfer an RF signal and single pulse with the same wavelength. Time and frequency signals are modulated in phase and intensity, respectively, so that they can be transferred with the same optical carrier. By this scheme, the channel resource can be saved in joint time and frequency transfer. A precise and continuous time and frequency transfer system based on the proposed scheme is demonstrated over a 110 km fiber link. Time and frequency signal are stabilized simultaneously owing to the ODL noise compensation. Results show the stability of frequency signal (ADEV) is $1.7\text{E} - 14/1\text{ s}$ and $5.9\text{E} - 17/10^4\text{ s}$, and the TDEV of time signal is $1.6\text{E} - 11/1\text{ s}$ and $9.1\text{E} - 13/10^4\text{ s}$. This scheme can be used for ultrastable frequency dissemination and time synchronization systems based on the metropolitan network.

Funding. Chinese Academy of Sciences (XDB21030200); National Natural Science Foundation of China (61535014).

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